

Paired Sawyer PickPlace rollout: ACT failure vs. REIM recovery

Meta-World/MuJoCo simulation frames · separate qualitative seed 8300042 · noise 0.2 · $\tau_{\text{on}} = 0.20$

(a) Paired start



ACT | $t=0$ | $d=0.244$ m

(b) Object displacement



ACT | $t=3$ | $\Delta=0.048$ m

(c) ACT drift



ACT | $t=199$ | $d=0.188$ m

(d) ACT failure



ACT | $t=200$ | $S=0$ | timeout

(e) Risk trigger



REIM | $t=9$ | $p=0.259$

(f) Recovery grasp / lift



REIM | $t=60$ | $z=0.052$ m

(g) Recovery transport



REIM | $t=61$ | $d=0.074$ m

(h) REIM success



REIM | $t=62$ | $S=1$ | $d=0.067$ m

Recovery control: until success or 150-step budget · orange: risk · green: recovery · simulated Sawyer frames